Verifiable Carbon Credits: MRV Infrastructure for Nature-Based Solutions

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Zoo Labs Foundation Research Paper ZOO-MRV-2026

With contributions from the Zoo Open Science Network

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RESEARCH

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Abstract

Nature-based carbon credits suffer a crisis of credibility. Recent analyses reveal that 70–90% of certified forest carbon offsets may represent phantom emissions reductions, undermining market integrity and climate goals. The root cause is inadequate Measurement, Reporting, and Verification (MRV) infrastructure that relies on infrequent manual audits, self-reported baselines, and opaque methodologies. We present **ZooMRV**, an open-source digital MRV platform that combines multi-spectral satellite imagery, LiDAR-derived biomass estimation, eddy covariance flux networks, and on-chain attestation to produce continuously verified carbon credit certificates. ZooMRV processes Sentinel-2, Landsat-9, and GEDI LiDAR data through a deep learning pipeline (CarbonNet) that estimates above-ground biomass with a root mean square error of 18.4 Mg/ha—a 34% improvement over the current operational standard. Credits are issued as non-fungible tokens on the Zoo Network with immutable provenance chains linking each credit to specific monitoring data, spatial boundaries, and verification reports. We demonstrate the system on 47 nature-based solution (NBS) projects across tropical forests, mangroves, peatlands, and grasslands, totaling 2.3 million hectares. ZooMRV detects 23% more reversals (biomass loss events) than traditional audit cycles and reduces the mean time between event and detection from 14 months to 12 days. The platform is governed through Zoo DAO processes and released under open licenses.

Keywords: carbon credits, MRV, satellite monitoring, biomass estimation, nature-based solutions, blockchain verification, remote sensing

Contents

1 Introduction

The voluntary carbon market (VCM) reached $2 billion in 2021 [Forest Trends, 2022], with nature-based solutions (NBS) representing the fastest-growing segment. NBS projects—including avoided deforestation (REDD+), reforestation, mangrove restoration, and peatland conservation—generate carbon credits by sequestering atmospheric CO_2 or preventing its release from terrestrial carbon stocks.

However, the scientific credibility of NBS carbon credits has been severely challenged. West et al. [2023] found that 94% of Verra's rainforest offset credits did not represent genuine emissions reductions. Guizar-Coutiño et al. [2022] showed that certified REDD+ projects in the Brazilian Amazon overestimated avoided deforestation by 1.8x on average. These findings have eroded buyer confidence and threaten the viability of carbon markets as a climate mitigation tool.

The fundamental problem is MRV infrastructure. Current practice relies on:

(i) **Infrequent verification**: Third-party audits occur every 2–5 years, leaving extended periods without monitoring [Schneider & La Hoz Theuer, 2019].

(ii) **Self-reported baselines**: Project developers construct their own counterfactual deforestation scenarios, creating systematic upward bias [West et al., 2023].

(iii) **Manual field sampling**: Biomass estimation relies on small plot samples extrapolated to vast areas, introducing large sampling errors [Chave et al., 2014].

(iv) **Opaque methodologies**: Verification reports are often proprietary, preventing independent scrutiny [Badgley et al., 2022].

(v) **Delayed detection**: Reversals (fire, illegal logging, drought mortality) may go undetected for years between audit cycles.

1.1 Digital MRV

Advances in satellite remote sensing, machine learning, and distributed ledger technology enable a fundamentally different approach to MRV—one that is continuous, transparent, and independently verifiable. Digital MRV (dMRV) replaces periodic manual audits with automated monitoring pipelines that process satellite data streams in near-real-time [Fuss et al., 2020].

1.2 Contributions

We present ZooMRV, an end-to-end digital MRV platform with the following contributions:

1. **CarbonNet**: A deep learning model for above-ground biomass (AGB) estimation from multi-spectral satellite imagery and LiDAR, achieving RMSE of 18.4 Mg/ha on independent validation data.

2. **Dynamic baselining**: An algorithmic approach to counterfactual construction using synthetic controls [Abadie et al., 2010] that eliminates self-reported baseline bias.

3. **Continuous monitoring**: Near-real-time change detection with 12-day median latency for reversal events, compared to 14 months for traditional audit cycles.

4. **On-chain attestation**: A blockchain-based credit registry where each credit is linked to verifiable satellite data, spatial boundaries, and algorithmic verification reports.

5. **Open validation**: All models, data, and methodologies are open-source, enabling independent verification and continuous improvement by the scientific community.

2 Background

2.1 Carbon Credit Standards

The VCM is dominated by four standards bodies: Verra (VCS), Gold Standard, American Carbon Registry, and Climate Action Reserve [Schneider & La Hoz Theuer, 2019]. Each maintains methodologies for quantifying emissions reductions and procedures for third-party verification. Despite these frameworks, systematic over-crediting has been documented across multiple project types [Badgley et al., 2022, West et al., 2023].

2.2 Remote Sensing for Forest Monitoring

Satellite-based forest monitoring has advanced dramatically with the Copernicus Sentinel program [Drusch et al., 2012] and NASA's GEDI mission [Dubayah et al., 2020]. Sentinel-2 provides 10 m resolution multi-spectral imagery every 5 days. GEDI provides spaceborne LiDAR measurements of forest canopy height and structure. Combined, these enable wall-to-wall biomass mapping at unprecedented spatial and temporal resolution [Lang et al., 2023].

2.3 Machine Learning for Biomass Estimation

Traditional allometric approaches estimate biomass from plot-level measurements of diameter, height, and wood density [Chave et al., 2014]. Machine learning approaches directly regress biomass from remote sensing features: random forests [Rodriguez-Veiga et al., 2021], gradient boosting [Hancock et al., 2019], and deep learning [Lang et al., 2023]. Recent models fusing optical, radar, and LiDAR data achieve continental-scale AGB maps with uncertainties of 20–30 Mg/ha [Xu et al., 2021].

2.4 Blockchain for Carbon Markets

Blockchain applications in carbon markets include credit tokenization [Schletz et al., 2020], registry interoperability [Franke et al., 2020], and automated retirement tracking [Howson, 2019]. Most existing platforms tokenize credits issued by traditional standards without addressing the underlying MRV quality problem. ZooMRV differs by generating native on-chain credits with embedded verification data.

3 CarbonNet: Biomass Estimation Model

3.1 Architecture

CarbonNet is a multi-modal deep learning model that fuses three data streams to estimate above-ground biomass:

1. **Optical branch**: A ResNet-50 encoder processes Sentinel-2 Level-2A surface reflectance imagery (13 spectral bands, 10–60 m resolution) through spatial and spectral attention.
2. **Radar branch**: A separate ResNet-50 encoder processes Sentinel-1 SAR backscatter (VV and VH polarizations, 10 m resolution), which penetrates cloud cover and is sensitive to woody biomass.
3. **LiDAR branch**: A point cloud encoder processes GEDI full-waveform LiDAR returns to extract canopy height profiles.

The three branches are fused through a cross-attention mechanism:

$$\mathbf{z} = \mathrm{CrossAttention}(\mathbf{h}_{\mathrm{opt}}, \mathbf{h}_{\mathrm{sar}}, \mathbf{h}_{\mathrm{lidar}}) \in \mathbb{R}^{256} \tag{1}$$

A regression head maps $\mathbf{z}$ to AGB in Mg/ha:

$$\hat{B} = \text{MLP}(\mathbf{z}) \in \mathbb{R}_{\geq 0} \tag{2}$$

3.2 Training Data

GEDI-Biomass training set. We use 4.2 million GEDI footprints (25 m diameter) with associated AGB estimates from the GEDI Level-4A product [Dubayah et al., 2022]. Each footprint is paired with co-located Sentinel-2 and Sentinel-1 imagery within a 30-day window.

Field plot calibration. We further calibrate against 12,847 field inventory plots from the Forest Observation System [Schepaschenko et al., 2019], spanning tropical, temperate, and boreal forests across 47 countries.

3.3 Loss Function

We minimize a composite loss:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda_1 \mathcal{L}_{\text{spatial}} + \lambda_2 \mathcal{L}_{\text{uncertainty}} \tag{3}$$

where $\mathcal{L}_{\text{MSE}}$ is the mean squared error on AGB, $\mathcal{L}_{\text{spatial}}$ is a spatial smoothness regularizer penalizing discontinuities at tile boundaries, and $\mathcal{L}_{\text{uncertainty}}$ is the negative log-likelihood of a heteroscedastic Gaussian that encourages well-calibrated uncertainty estimates.

3.4 Results

Table 1: AGB estimation accuracy on independent validation plots. RMSE in Mg/ha. Best results in **bold**.

Model	Tropical	Temperate	Boreal	All
Allometric (plot-based)	42.1	31.7	24.3	35.2
Random Forest	31.4	24.8	19.1	27.6
GBM (XGBoost)	29.8	23.1	18.4	25.9
ESA CCI Biomass	33.7	25.6	20.7	28.3
Lang et al. (2023)	26.2	20.4	16.8	22.1
CarbonNet	**22.7**	**17.3**	**13.8**	**18.4**

CarbonNet achieves 18.4 Mg/ha RMSE across all biomes, a 34% improvement over the current operational standard (ESA CCI Biomass). The improvement is largest in tropical forests (+33%), where dense canopy structure makes biomass estimation most challenging.

4 Dynamic Baselining

4.1 The Baseline Problem

Carbon credits quantify the difference between actual outcomes and a counterfactual baseline—what would have happened without the intervention. Traditional approaches allow project developers to construct their own baselines, creating a fundamental conflict of interest. Higher baseline deforestation projections yield more credits, incentivizing inflated baselines [West et al., 2023].

4.2 Synthetic Control Approach

We adopt the synthetic control method [Abadie et al., 2010] for baseline construction. For each project area P, we identify a set of J control areas $\{C_1, \ldots, C_J\}$ that matched the project area's pre-intervention characteristics but were not subject to conservation intervention.

The synthetic control baseline is:

$$\hat{B}_t^{\text{baseline}}(P) = \sum_{j=1}^{J} w_j \cdot B_t(C_j) \tag{4}$$

where weights $w_j \geq 0$, $\sum w_j = 1$ are chosen to minimize pre-intervention prediction error:

$$\mathbf{w}^* = \arg\min_{\mathbf{w}} \sum_{t<t_0} \left| B_t(P) - \sum_j w_j B_t(C_j) \right|^2 \tag{5}$$

4.3 Control Area Selection

Control areas are selected from a candidate pool using the following matching criteria:

- Same biome and eco-region (WWF classification).
- Similar deforestation pressure (road density, population growth, commodity prices).
- Similar forest type and initial biomass (±20%).
- No conservation intervention during the study period.
- Minimum 50 km buffer from the project boundary.

4.4 Baseline Validation

We validate dynamic baselines against actual deforestation in 23 projects where the true counterfactual is observable (projects that lost funding and were abandoned). Dynamic baselines predict actual deforestation with $R^2 = 0.87$, compared to $R^2 = 0.41$ for project-developer baselines and $R^2 = 0.72$ for jurisdictional averages.

5 Continuous Monitoring Pipeline

5.1 Data Ingestion

ZooMRV ingests data from multiple satellite constellations:

Table 2: Satellite data sources and processing characteristics.

Source	Resolution	Revisit	Coverage	Primary use
Sentinel-2	10 m	5 days	Global	AGB, change detection
Sentinel-1 SAR	10 m	6 days	Global	Cloud-free monitoring
Landsat-9	30 m	16 days	Global	Long-term trends
GEDI LiDAR	25 m	—	Tropical	Canopy height, AGB cal.
Planet NICFI	4.7 m	Monthly	Tropical	Fine-scale change

5.2 Change Detection

We implement a multi-scale change detection algorithm:

Pixel-level. A temporal convolutional network (TCN) processes per-pixel time series of NDVI, NBR, and SWIR2 bands to detect abrupt changes (logging, fire) and gradual degradation:

$$\hat{c}_i = \mathrm{TCN}(\mathbf{x}_{i,t-T:t}) \in [0, 1] \tag{6}$$

where $\hat{c}_i$ is the change probability for pixel i.

Object-level. A U-Net semantic segmentation model identifies changed areas at the patch level, providing spatial context that reduces false positives from atmospheric and phenological variation.

Alert aggregation. Pixel and object-level detections are aggregated into alert polygons with associated confidence scores, change type classification (deforestation, degradation, fire, flood), and estimated biomass loss.

5.3 Reversal Detection Performance

Table 3: Reversal detection performance across ecosystem types.

Ecosystem	Detection rate	False positive rate	Median latency
Tropical forest	94.2%	3.1%	8 days
Mangrove	91.7%	4.8%	11 days
Peatland	87.3%	6.2%	18 days
Grassland/savanna	83.6%	8.1%	15 days
Overall	**91.4%**	**4.7%**	**12 days**

ZooMRV detects 91.4% of reversal events with a median latency of 12 days, compared to the 14-month average for traditional audit cycles. The system detects 23% more reversals than periodic audits, primarily small-scale degradation events that fall below audit detection thresholds.

6 On-Chain Credit Registry

6.1 Credit Lifecycle

Each carbon credit follows a lifecycle managed by smart contracts:

1. **Project registration**: Project boundaries, methodology, and baseline parameters are recorded on-chain.
2. **Monitoring**: CarbonNet biomass estimates and change detection results are committed as merkle roots at each monitoring interval.
3. **Issuance**: Credits are minted as ERC-721 tokens when verified carbon stock gains exceed the dynamic baseline by a statistically significant margin.
4. **Buffer pool**: 20% of issued credits are held in a permanence buffer pool, released over the project crediting period if no reversals occur.

5. **Retirement**: Credits are permanently burned when used for offsetting claims, with the retirement event linked to the retiree's account.
6. **Reversal adjustment**: If reversals are detected, buffer pool credits are automatically cancelled proportional to the biomass loss.

6.2 Credit Metadata

Each credit token contains or references:

- Project ID and spatial boundary (GeoJSON hash).
- Vintage year and monitoring period.
- Biomass estimate with uncertainty bounds.
- Dynamic baseline value.
- Net sequestration (credit quantity in tCO_2e).
- Links to raw satellite imagery (IPFS CIDs).
- CarbonNet model version and parameters.
- Reversal risk score.

6.3 Transparency and Auditability

Any party can independently verify a credit by:

1. Downloading the referenced satellite imagery from IPFS.
2. Running the open-source CarbonNet model.
3. Comparing the result against the on-chain biomass attestation.
4. Verifying the dynamic baseline against published control areas.

This creates a fully reproducible verification chain—a fundamental departure from the opaque auditor reports of traditional carbon standards.

7 Evaluation on NBS Projects

7.1 Project Portfolio

We evaluate ZooMRV on 47 nature-based solution projects:

Table 4: Project portfolio by ecosystem type.

Ecosystem	**Projects**	**Area (Mha)**	**Traditional credits**	**ZooMRV credits**
Tropical forest	21	1.42	18.7M	12.4M
Mangrove	8	0.23	2.1M	1.8M
Peatland	7	0.38	4.3M	2.9M
Grassland/savanna	6	0.18	1.4M	1.1M
Temperate forest	5	0.12	0.9M	0.8M
Total	**47**	**2.33**	**27.4M**	**19.0M**

7.2 Over-Crediting Analysis

ZooMRV identifies systematic over-crediting in the traditional portfolio:

- **Inflated baselines**: Dynamic baselines reduce creditable deforestation avoidance by 31% on average, with the largest corrections in tropical forest REDD+ projects (37%).
- **Undetected reversals**: 23% more reversal events detected by continuous monitoring, reducing net credits by 8%.
- **Biomass overestimation**: CarbonNet corrects systematic upward bias in allometric estimates, reducing per-hectare credits by 12% in dense tropical forests.

Overall, ZooMRV issues 30.7% fewer credits than traditional certification for the same project portfolio. This reduction represents phantom credits that do not correspond to real climate benefit.

7.3 Uncertainty Quantification

CarbonNet provides pixel-level uncertainty estimates. At the project level, we report:

Table 5: Credit uncertainty by ecosystem type (95% CI as % of mean).

Ecosystem	ZooMRV uncertainty	Traditional uncertainty
Tropical forest	±14%	±28%
Mangrove	±18%	±35%
Peatland	±22%	±42%
Grassland	±19%	±38%

ZooMRV reduces credit uncertainty by approximately 50% across all ecosystem types, primarily through the fusion of multiple remote sensing data sources and continuous temporal sampling.

8 Discussion

8.1 Market Implications

The 30.7% over-crediting correction has significant market implications. If applied across the entire VCM, approximately 200 million credits out of the 650 million issued for NBS projects may not represent genuine climate benefit. This supports the growing consensus that quality assurance—not volume growth—is the primary challenge for carbon markets [Badgley et al., 2022].

However, fewer high-quality credits command premium prices. The voluntary market is increasingly stratified, with "premium" credits verified by advanced dMRV commanding 2–5x price premiums over standard credits. This price signal can drive adoption of ZooMRV-class verification infrastructure.

8.2 Advantages of Digital MRV

1. **Transparency**: Open-source models and publicly verifiable data chains eliminate the "black box" problem of traditional verification.
2. **Scalability**: Satellite-based monitoring scales to millions of hectares without proportional cost increases.

3. **Timeliness**: Near-real-time reversal detection enables rapid response and market-responsive credit adjustment.

4. **Consistency**: Algorithmic baselines and verification eliminate auditor variability and subjective judgment.

5. **Cost efficiency**: After initial infrastructure investment, per-credit verification costs are estimated at $0.02–0.05, compared to $0.50–2.00 for traditional audits.

8.3 Limitations

1. **Below-canopy changes**: Satellite-based monitoring cannot detect forest degradation that occurs beneath the canopy (selective logging of understory, biodiversity loss). Acoustic and camera trap monitoring can complement.

2. **Soil carbon**: CarbonNet models above-ground biomass only. Soil organic carbon, which represents the majority of carbon in peatlands and grasslands, requires separate modeling approaches.

3. **Cloud cover**: Persistent cloud cover in tropical regions reduces optical satellite data availability. Sentinel-1 SAR provides all-weather coverage but with lower sensitivity to biomass.

4. **GEDI data gaps**: GEDI's sampling pattern leaves gaps that are filled by extrapolation, introducing uncertainty in areas without direct LiDAR measurements.

5. **Additionality**: Dynamic baselines address over-crediting but do not fully resolve the fundamental challenge of proving that conservation would not have occurred without carbon finance.

9 Implementation

9.1 Software Architecture

ZooMRV is implemented as a set of containerized microservices:

- **Data ingestion**: Google Earth Engine and Planetary Computer APIs for satellite data access; custom GEDI processor.

- **CarbonNet inference**: PyTorch 2.2 model serving on NVIDIA A100 GPUs; batch processing of Sentinel-2 tiles.

- **Change detection**: Temporal analysis pipeline using NumPy, Rasterio, and scikit-image.

- **Baseline engine**: Synthetic control computation using SciPy optimization; control area matching with H3 spatial indexing.

- **Smart contracts**: Solidity contracts on Zoo Network; IPFS for data storage.

- **API layer**: FastAPI REST API exposing project status, credit issuance, and verification endpoints.

9.2 Computational Requirements

Processing a 1-million-hectare project requires approximately:

- 2.4 GPU-hours for CarbonNet inference per monitoring period.
- 0.8 CPU-hours for change detection per monitoring period.
- 0.2 CPU-hours for baseline computation (one-time setup).
- 12 GB storage per monitoring period (compressed imagery and derived products).

Annual monitoring of the full 2.3 Mha portfolio costs approximately $18,000 in compute, or $0.008 per hectare—orders of magnitude cheaper than field-based monitoring.

10 Future Work

1. **Soil carbon modeling**: Extending CarbonNet with soil organic carbon estimation using microwave remote sensing and thermal infrared data.
2. **Biodiversity co-benefits**: Integrating TaxoNet species monitoring (ZOO-CLS-2026) to verify biodiversity additionality alongside carbon sequestration.
3. **Blue carbon**: Specialized models for mangrove, seagrass, and salt marsh ecosystems using SAR and bathymetric data.
4. **Decentralized processing**: Distributing CarbonNet inference across the Zoo compute network, enabling community-operated verification nodes.
5. **Integration with BIBs**: Linking ZooMRV verification to Biodiversity Impact Bond (ZOO-BIB-2026) oracle networks for automated outcome assessment.
6. **Article 6 compliance**: Aligning credit issuance with Paris Agreement Article 6 corresponding adjustment requirements for international transfers.

11 Conclusion

Verifiable carbon credits require verifiable infrastructure. ZooMRV demonstrates that continuous satellite-based monitoring, algorithmic baselining, and on-chain attestation can produce carbon credits with substantially higher integrity than traditional certification. The 30.7% over-crediting correction identified in our evaluation portfolio underscores the urgency of MRV modernization. By reducing verification costs by orders of magnitude while increasing accuracy and transparency, digital MRV can restore confidence in nature-based carbon markets and channel genuine finance toward climate mitigation.

Through Zoo Labs Foundation's commitment to open science, ZooMRV's models, data pipelines, and smart contracts are freely available. The platform is governed through Zoo DAO processes, ensuring that methodology updates, buffer pool parameters, and credit issuance standards evolve through community consensus rather than proprietary decision-making.

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